

AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation

PHASE I REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Tourism planning often requires travelers to manually search for tourist attractions, identify unfamiliar landmarks, locate nearby facilities, estimate suitable visiting times, and prepare travel schedules using multiple applications. This project presents an AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation to provide these functionalities through a unified mobile application. The proposed system enables users to capture or upload an image of a tourist landmark, which is analyzed using a deep learning model based on MobileNetV2 and transfer learning to identify the corresponding tourist destination. The trained landmark recognition model is converted into TensorFlow Lite (TFLite) format for efficient deployment within a Flutter mobile application, enabling on-device image classification without depending on paid cloud-based image recognition services.

Following landmark identification, the system obtains geographical information such as latitude and longitude using open geolocation services. OpenStreetMap-based services are utilized to identify nearby tourist attractions, hotels, restaurants, and other relevant points of interest. Based on the recognized destination, number of travel days, and available attractions, the system generates a structured day-wise personalized itinerary containing suggested places and suitable visiting schedules. Map integration enables users to visualize destinations and access travel directions. The application also provides a travel journal facility that allows users to store visited places, photographs, and personal travel experiences. Hotel information and external booking options can additionally assist users in preparing their trips. By integrating deep learning, mobile computing, geolocation, map services, and intelligent itinerary planning, the proposed system provides a convenient and unified platform that reduces manual travel planning and assists users in discovering, organizing, and documenting their tourism experiences.

Keywords: Artificial Intelligence, Smart Tourism, Landmark Recognition, MobileNetV2, Transfer Learning, TensorFlow Lite, Flutter, Geolocation, OpenStreetMap, Personalized Itinerary, Tourism Recommendation, Travel Journal.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
POI	Point of Interest
RAM	Random Access Memory
SDK	Software Development Kit
TFLite	TensorFlow Lite
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Tourism has become an important part of modern society, providing opportunities for people to explore historical sites, cultural landmarks, natural attractions, entertainment destinations, and local experiences. With the growth of smartphones and digital services, travelers increasingly depend on mobile applications to search for destinations, obtain geographical information, find nearby attractions, and organize their trips. However, tourism-related information is often distributed across different applications and websites, requiring users to perform multiple searches before completing their travel plans.

Artificial Intelligence (AI) and deep learning techniques have created new opportunities for developing intelligent tourism applications. Image recognition can be used to identify tourist landmarks from photographs, while geolocation services can provide the geographical coordinates of recognized destinations. Location-based services can further help users discover nearby attractions, hotels, restaurants, and other points of interest. Combining these technologies can provide travelers with more convenient and context-aware tourism assistance.

The proposed AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation is designed as a mobile application that integrates landmark recognition, geographical information, nearby-place discovery, map services, and itinerary planning. The system allows users to capture or upload an image of a tourist landmark and uses a MobileNetV2-based transfer learning model to identify the landmark. The trained model is converted into TensorFlow Lite (TFLite) format for efficient deployment within the Flutter mobile application.

After recognizing the landmark, the application retrieves its geographical location and uses OpenStreetMap-based services to identify nearby tourist attractions, hotels, restaurants, and other useful locations. Based on the destination and selected travel duration, the application generates a structured day-wise itinerary. Map and navigation facilities assist users in locating destinations, while the travel journal allows them to record visited places, photographs, and personal experiences. Thus, the proposed system combines AI, mobile computing, geolocation, mapping, and intelligent travel planning into a unified tourism platform.

1.2 OBJECTIVE

The main objective of this project is to develop an AI-powered mobile tourism planner that simplifies destination identification and travel schedule preparation through an integrated application.

The specific objectives are:

1. To develop a landmark recognition system using MobileNetV2 and transfer learning for identifying tourist destinations from uploaded or captured images.
2. To convert the trained landmark recognition model into TensorFlow Lite format and integrate it into a Flutter mobile application for efficient on-device inference.
3. To retrieve geographical information such as latitude and longitude for the recognized tourist landmark using open geolocation services.
4. To identify nearby tourist attractions, hotels, restaurants, and other relevant points of interest using OpenStreetMap-based services.
5. To generate a personalized day-wise itinerary based on the selected destination, available attractions, and number of travel days.
6. To integrate interactive maps and directions to help users locate and navigate to selected tourism destinations.
7. To provide a travel journal facility for storing visited places, photographs, and personal travel experiences.
8. To provide hotel information and external booking options to support users during travel preparation.

1.3 EXISTING SYSTEM

Existing tourism planning generally depends on multiple websites and mobile applications for completing different travel-related activities. Users may use one application to identify or search for a landmark, another application to locate nearby attractions, a separate map application for directions, and different platforms for hotels, restaurants, and booking information. This fragmented approach requires users to repeatedly search for information and manually combine the obtained details into a travel plan.

Most conventional tourism applications primarily provide destination information, maps, reviews, or recommendations individually rather than integrating landmark recognition with complete itinerary generation. In addition, preparing a multi-day travel schedule usually requires

the user to manually select attractions, determine their order, estimate visiting times, and distribute them across different days.

Cloud-based image recognition services can also be used for identifying landmarks, but their use may involve internet dependency, API restrictions, usage limits, or additional service costs. Such approaches may not be suitable when an application requires efficient and direct landmark recognition within a mobile environment.

Disadvantages of the Existing System

- Dependence on multiple tourism applications and websites.
- Manual identification and searching of tourist destinations.
- Manual preparation of day-wise travel schedules.
- Limited integration between landmark recognition and tourism planning.
- Separate searching for nearby attractions, hotels, and restaurants.
- Dependence on external or cloud-based image recognition services in some systems.
- Difficulty in organizing discovered tourism information into a structured itinerary.
- Limited integration of travel planning and personal travel documentation.

1.4 PROPOSED SYSTEM

The proposed system, AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation, provides an integrated mobile platform for destination identification and travel planning. The system is developed using Flutter and combines deep learning, geolocation, OpenStreetMap services, map integration, and itinerary generation within a single application.

The user can capture a photograph using the mobile camera or upload an existing landmark image. The image is preprocessed and passed to a MobileNetV2-based transfer learning model trained to recognize selected tourist landmarks. The trained model is converted into TensorFlow Lite format, allowing the model to perform landmark classification efficiently within the mobile application.

Once the landmark is recognized, the system obtains the corresponding geographical coordinates through open geolocation services. These coordinates are used with OpenStreetMap-based services to discover nearby tourist attractions, hotels, restaurants, and other points of interest. The user can explore these locations through an interactive map and obtain directions to selected destinations.

The system also generates a personalized day-wise itinerary according to the destination, number of travel days, and available attractions. The itinerary organizes recommended places into a structured travel schedule to reduce the effort required for manual planning. In addition, hotel information and external booking options can be provided to assist users in preparing their accommodation arrangements.

A travel journal module allows users to record their tourism experiences by saving visited locations, photographs, and personal notes. The overall system follows the workflow:

*Image Capture/Upload → Image Preprocessing → Landmark Recognition →
Geolocation → Nearby Place Discovery → Personalized Itinerary → Map &
Directions → Travel Journal*

The proposed system therefore provides a unified tourism planning environment that reduces the need to switch between multiple applications and supports users throughout the destination discovery, planning, navigation, and travel documentation process.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The literature review for the proposed AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation focuses on recent developments in artificial intelligence, landmark recognition, tourism recommendation, geolocation, mobile computing, and automated itinerary generation.

Recent tourism applications increasingly use artificial intelligence and computer vision to improve destination discovery and personalization. Deep learning-based image classification techniques enable systems to recognize landmarks from photographs and provide users with relevant destination information. Transfer learning is particularly useful for landmark recognition because pretrained convolutional neural networks can extract meaningful visual features while reducing training requirements.

Lightweight architectures such as MobileNetV2 are suitable for mobile applications because they provide efficient image classification with relatively lower computational requirements. The conversion of trained models into TensorFlow Lite format further enables on-device inference, reducing dependence on remote cloud-based image recognition services.

Location-based tourism systems use geographical coordinates to identify attractions and services surrounding a destination. OpenStreetMap and related open geospatial services provide information about points of interest such as tourist attractions, restaurants, hotels, and other facilities. These services can be integrated with mobile applications to provide interactive maps and location-based discovery.

Tourism recommendation systems have also been developed to recommend destinations according to user preferences, location, available time, and travel requirements. Automated itinerary generation extends this concept by organizing selected attractions into a structured schedule over multiple days. However, many existing approaches focus on individual aspects such as recommendation, navigation, or landmark recognition rather than combining these functions into a single mobile tourism platform.

The proposed system addresses this integration gap by combining landmark recognition, geolocation, nearby-place discovery, personalized itinerary generation, map services, hotel information, and travel journaling within a unified Flutter application.

Recent research on smart tourism falls mainly into three groups: tourism recommendation systems, itinerary planning and optimization, and image-based tourist attraction or landmark recognition. The reviewed works are summarized in Table 2.1.

Table 2.1. Summary of the reviewed literature

Ref.	Focus of the work	Relevance to the proposed system
[1]	Hybrid, content-boosted collaborative filtering for a smart tourism recommender (2024).	Shows recommendation techniques for tourists; does not include landmark recognition or itinerary scheduling.
[2]	Two-stage travel itinerary optimization considering stochastic traffic time (2024).	Supports the itinerary-generation module; no image-based destination identification.
[3]	Recommendation rules to personalize itineraries for tourists in an unfamiliar city (2024).	Relevant to personalized itineraries for first-time visitors.
[4]	Adaptive tourism recommendation system (2024).	Relevant to adapting plans to the traveler; no on-device recognition.
[5]	Deep reinforcement learning to recommend sequences of attractions for sustainable tourism (2025).	Relevant to ordering attractions within a trip.
[6]	Multi-day itinerary recommendation using multi-source online data (2024).	Directly relevant to the day-wise itinerary module.
[7]	Personalized recommendation using a temporal multilayer sequential neural network (2025).	Shows deep learning for tourism recommendation.
[8]	+Tour: personalized itineraries for smart tourism (2025).	Relevant to smart-tourism itinerary generation.
[9]	Deep learning for personalized recommendation, behavior analysis and route optimization (2025).	Relevant to route planning and personalization.
[10]	T-ECBM: text-image multimodal model for tourist attraction recommendation (2025).	Uses image data for tourism, as the proposed system does.
[11]	Multi-objective contextual bandits for smart tourism recommendation (2025).	Context-aware recommendation approach.
[12]	Deep neural associative recommendation with real-time context injection (2025).	Relevant to real-time, context-aware suggestions.
[13]	Tourist attraction image recognition and intelligent recommendation using deep learning (2025).	Closest to the landmark recognition module.
[14]	Ensemble deep learning for landmark recognition using pixel-enhanced YOLO11 models (2025).	Landmark recognition; the proposed system uses a lightweight MobileNetV2 for on-device use.
[15]	MCR-TRec: multi-criteria tourism recommendation for Tamil Nadu using NLP (2025).	Regional tourism recommendation.

Recommendation and itinerary studies [1]–[12], [15] show that deep learning, sequential models, reinforcement learning and context-aware methods can generate personalized travel plans. However, these works take the destination as given and do not identify it from a photograph.

Image-based studies [10], [13], [14] show that deep learning can recognize tourist attractions from images. These works concentrate on recognition accuracy or recommendation and do not combine recognition with nearby-place discovery, day-wise itinerary generation and a travel journal in one mobile application.

2.2 COMPARISON AND DISCUSSION

The proposed system combines multiple tourism-related functions that are commonly provided separately by existing systems.

Table 2.2. Comparison of traditional tourism applications and the proposed tourism planner

Feature	Traditional Tourism Applications	Proposed Tourism Planner
Landmark Recognition	Limited / Separate Service	✓
Image-based Destination Identification	Limited	✓
Transfer Learning	Application dependent	✓
MobileNetV2	Not commonly integrated	✓
TFLite On-device Recognition	Limited	✓
Geolocation	✓	✓
Tourist Attraction Search	✓	✓
Nearby Hotels	✓	✓
Nearby Restaurants	✓	✓
Interactive Maps	✓	✓
Personalized Itinerary	Limited	✓
Day-wise Schedule	Manual / Limited	✓
Travel Journal	Limited	✓
Unified Mobile Platform	Limited	✓

The comparison shows that conventional tourism applications generally provide destination information, maps, or booking facilities individually. The proposed system integrates these functions with an AI-based landmark recognition component and itinerary generation

mechanism. This integration reduces the need for users to manually collect information from multiple applications.

2.3 CONCLUSION

The literature survey indicates that artificial intelligence, deep learning, computer vision, geolocation, and mobile computing can significantly support modern tourism applications. Landmark recognition can simplify destination identification, while location-based services can provide information about nearby attractions and facilities. Transfer learning and lightweight architectures such as MobileNetV2 are suitable for developing efficient image recognition models for mobile environments.

Existing systems demonstrate the usefulness of individual tourism functions, but there is scope for integrating landmark recognition, geographical discovery, itinerary generation, navigation, and travel documentation into a single application. Therefore, the proposed AI-Based Smart Tourism Planner combines these technologies to provide an integrated platform for tourism discovery and personalized trip planning.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed system follows a modular architecture consisting of the Flutter mobile application, landmark recognition model, geolocation services, OpenStreetMap-based location services, itinerary generation component, map integration, and travel journal.

The overall architecture of the proposed system is shown in Figure 3.1.

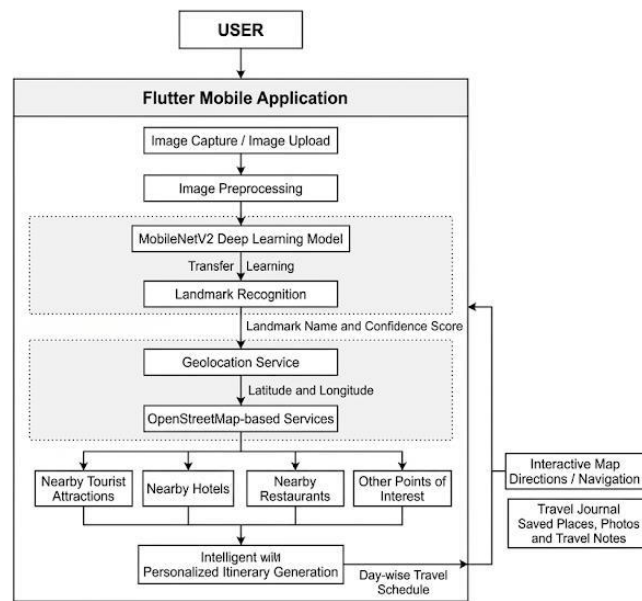


Figure 3.1. Architecture diagram of the proposed system

3.2 SYSTEM REQUIREMENTS

The system requirements define the software, hardware, dataset, and deployment resources required to develop and operate the proposed tourism planner.

3.2.1 SOFTWARE REQUIREMENTS

Table 3.1. Software requirements of the proposed system

Requirement	Purpose
Flutter SDK	Mobile application development
Dart	Flutter application programming
Python	AI model development and preprocessing
TensorFlow/Keras	Deep learning model development
MobileNetV2	Landmark image classification
TensorFlow Lite	Mobile model deployment
OpenStreetMap	Map and POI information
Geolocation Service	Latitude and longitude retrieval
Android Studio / VS Code	Application development
Android SDK	Android application deployment
Database / Local Storage	Travel journal and user information

3.2.2 HARDWARE REQUIREMENTS

The project does not require specialized physical hardware. A standard development computer and an Android smartphone are sufficient.

Table 3.2. Hardware requirements of the proposed system

Hardware	Recommended Requirement
Processor	Intel Core i5 / AMD equivalent or above
RAM	8 GB or above
Storage	20 GB or more available
Smartphone	Android smartphone
Camera	Smartphone camera
Internet	Required for online map/location services

3.2.3 DATASET REQUIREMENTS

The landmark recognition model requires a labeled image dataset containing multiple tourist landmark categories. Each image must be associated with the correct landmark class.

The dataset preparation process consists of:

Landmark Image Dataset → Image Collection → Data Cleaning → Duplicate/Invalid Image Removal → Image Resizing → Normalization → Data Augmentation → Training Dataset / Validation Dataset / Testing Dataset

The dataset requirements will include:

- Multiple landmark categories.
- Sufficient images for each category.
- Correct class labels.
- Different viewing angles.
- Different lighting conditions.
- Different image backgrounds.
- Training, validation, and testing subsets.
- Consistent image dimensions.
- Properly balanced classes wherever possible.

3.2.4 DEPLOYMENT AND SCALING

The trained landmark recognition model is converted into TensorFlow Lite format for deployment in the Flutter mobile application. This allows the application to perform landmark prediction directly on the mobile device.

The deployment process is:

Trained MobileNetV2 Model → TensorFlow/Keras Model → TFLite Conversion → .tflite Model → Flutter Assets → Mobile Application → On-device Landmark Prediction

The application can be scaled by:

- Adding additional landmark classes.
- Updating the landmark dataset.
- Improving the trained model.
- Adding additional tourism APIs.
- Supporting more geographical regions.
- Adding multilingual functionality.
- Supporting more user preferences.
- Introducing cloud/backend services if required.

3.3 SYSTEM DESIGN

The system design describes the interaction between the user, AI model, location services, tourism information services, itinerary generator, and mobile application.

The major system components are:

1. User Interface
2. Image Processing
3. Landmark Recognition
4. Geolocation
5. Nearby Place Discovery
6. Itinerary Generation
7. Map and Navigation
8. Travel Journal

3.3.1 ACTIVITY DIAGRAM

The activity diagram in Figure 3.2 represents the complete operation from opening the application to creating and saving a travel plan.

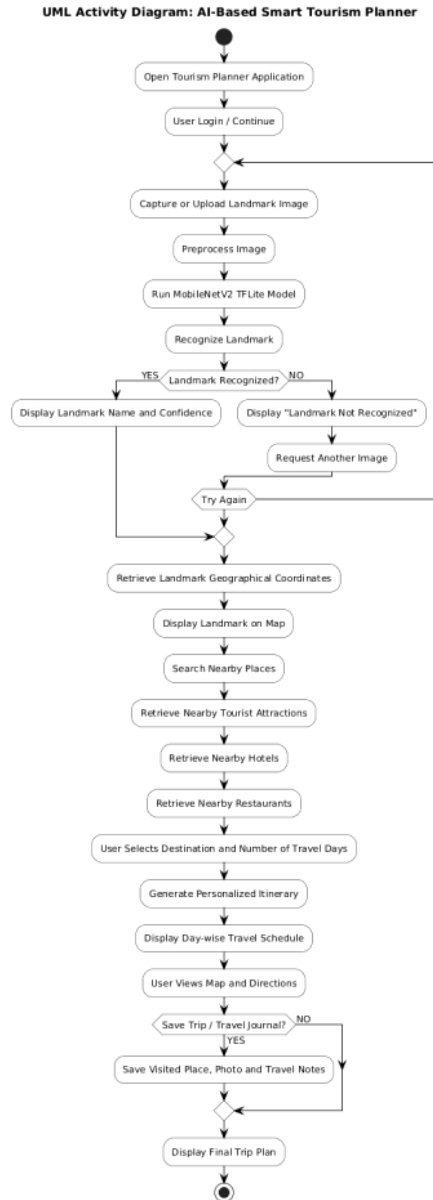


Figure 3.2. Activity diagram of the proposed system

3.3.2 DATA FLOW DIAGRAM

The data flow diagram of the proposed system is shown in Figure 3.3.

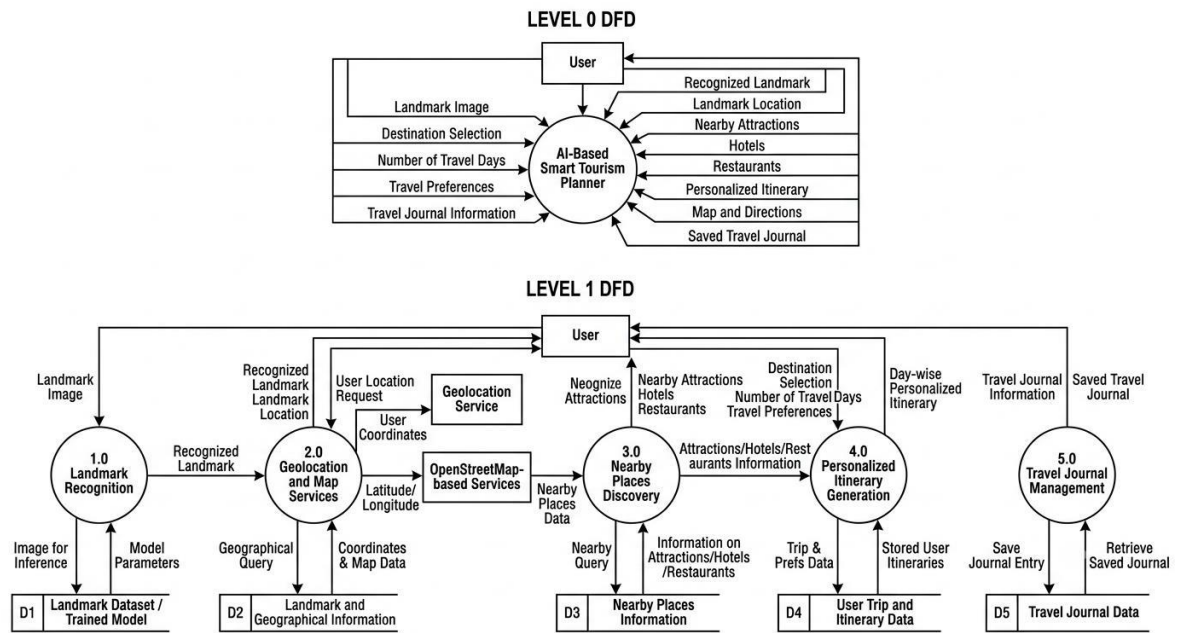


Figure 3.3. Data flow diagram of the proposed system

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The proposed AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation integrates deep learning, mobile application development, geolocation, map services, and intelligent travel planning. The system is designed to identify tourist landmarks from images and provide relevant destination information, nearby places, and a structured travel itinerary through a unified mobile application.

The major methodologies used in the proposed system are described below.

4.1.1 IMAGE PRE-PROCESSING

The input landmark image is first prepared before being provided to the deep learning model. Images are resized to the input dimensions required by the MobileNetV2 model and normalized to obtain a consistent numerical representation. Data augmentation techniques can be applied during training to improve model generalization under different image conditions.

4.1.2 TRANSFER LEARNING

Transfer learning is used to develop the landmark recognition model. Instead of training a deep convolutional neural network completely from the beginning, a pretrained MobileNetV2 network is utilized as the feature extraction backbone. The final classification layers are adapted to recognize the landmark categories available in the tourism dataset. This approach reduces training requirements and enables effective learning from a relatively limited domain-specific dataset.

4.1.3 MobileNetV2-BASED LANDMARK RECOGNITION

MobileNetV2 is selected as the primary image classification architecture because of its lightweight design and suitability for mobile applications. The model extracts visual features from the input image and produces class probabilities corresponding to the trained tourist landmarks. The landmark with the highest predicted probability is selected as the recognition result.

4.1.4 TENSORFLOW LITE DEPLOYMENT

After model training and evaluation, the trained model is converted into TensorFlow Lite format. The TFLite model is integrated into the Flutter application so that landmark prediction can be performed on the mobile device. This reduces dependence on a continuously running cloud-based image recognition service for the core landmark classification task.

4.1.5 GEOLOCATION-BASED INFORMATION RETRIEVAL

After the landmark is recognized, its geographical information is obtained through an open geolocation service. Latitude and longitude values are used to locate the destination on the map and serve as the reference point for nearby-place searches.

4.1.6 OpenStreetMap-BASED PLACE DISCOVERY

OpenStreetMap-based services are used to retrieve relevant points of interest around the recognized destination. The system can identify tourist attractions, hotels, restaurants, and other useful locations based on geographical proximity.

4.1.7 PERSONALIZED ITINERARY GENERATION

The itinerary generation methodology organizes available attractions into a day-wise travel schedule. The destination, number of travel days, and available places are considered while preparing the itinerary. The resulting schedule provides users with an organized sequence of places to visit during their trip.

4.1.8 MOBILE APPLICATION INTEGRATION

Flutter is used to combine all major functionalities into a single mobile application. The application provides interfaces for landmark recognition, destination information, maps, nearby places, itinerary planning, travel journal management, and related travel information.

4.2 MODULES

The proposed system is divided into five major modules:

1. Landmark Recognition Module
2. Geolocation and Map Module
3. Nearby Places Module
4. Personalized Itinerary Module
5. Travel Journal and Booking Module

These modules work together to provide a complete tourism planning workflow.

4.2.1 LANDMARK RECOGNITION MODULE

The Landmark Recognition Module is the primary AI component of the proposed system. It enables users to identify a tourist destination from an image captured using the mobile camera or selected from the device gallery.

The image is first passed through preprocessing operations such as resizing and normalization. The processed image is then provided to the MobileNetV2-based transfer learning model. The model extracts visual features and classifies the image into one of the trained landmark categories.

Processing Flow

Image Capture / Upload → Image Pre-processing → Image Resizing → Normalization → MobileNetV2 → Feature Extraction → Classification → Landmark Name + Confidence

The trained model is converted into TFLite format and integrated into the Flutter application. When the user provides an image, the TFLite interpreter performs inference and returns the predicted landmark.

The module provides:

- Image capture through the mobile camera.
- Image selection from device storage.
- Image preprocessing.
- Landmark classification.
- Prediction confidence.
- Recognized landmark name.
- TFLite-based mobile inference.

The output of this module is passed to the Geolocation and Map Module for obtaining destination information.

4.2.2 GEOLOCATION AND MAP MODULE

The Geolocation and Map Module provides geographical information for the recognized landmark. Once the landmark has been identified, the system retrieves its corresponding geographical coordinates, including latitude and longitude.

These coordinates are used to display the landmark on an interactive map. The module acts as the connection between landmark recognition and location-based tourism services.

Processing Flow

Recognized Landmark → Geolocation Service → Latitude + Longitude → Map Location → Destination Marker → Directions / Navigation

The module provides:

- Landmark location.
- Latitude and longitude.
- Interactive map display.
- Destination markers.
- Location visualization.
- Route/direction support where implemented.

The geographical coordinates obtained by this module are also supplied to the Nearby Places Module for discovering tourism facilities around the destination.

4.2.3 NEARBY PLACES MODULE

The Nearby Places Module helps users discover useful locations surrounding the recognized tourist destination. OpenStreetMap-based services are utilized to retrieve points of interest according to the destination's geographical coordinates.

The module primarily focuses on:

- Tourist attractions.
- Hotels.
- Restaurants.
- Other relevant points of interest.

The application presents the retrieved places with relevant location information. Users can select individual places and view them on the map. The information obtained from this module forms an important input for the Personalized Itinerary Module.

4.2.4 PERSONALIZED ITINERARY MODULE

The Personalized Itinerary Module is responsible for preparing a structured travel schedule based on the user's selected destination and travel duration.

The user provides the required number of travel days, and the system uses the available tourist attractions to create a day-wise schedule. Places can be organized according to their relevance and geographical proximity to make the generated plan easier to follow.

Processing Flow

Destination + Number of Travel Days + Available Attractions → Itinerary Generation → Day-wise Organization → Morning / Afternoon / Evening → Personalized Travel Schedule

For example:

DAY 1

Morning → Attraction A

Afternoon → Attraction B

Evening → Attraction C

DAY 2

Morning → Attraction D

Afternoon → Attraction E

Evening → Attraction F

The module reduces the effort involved in manually selecting and arranging tourist destinations. The generated itinerary can be displayed together with map information so that users can understand the location of each recommended place.

The module considers:

- Selected destination.
- Number of travel days.
- Available tourist attractions.
- Place information.
- Visiting sequence.
- Day-wise distribution.

4.2.5 TRAVEL JOURNAL AND BOOKING MODULE

The Travel Journal and Booking Module provides additional facilities for managing the user's travel experience. The travel journal allows users to maintain records of places they have visited and store personal travel information.

Users can maintain:

- Visited place names.
- Travel photographs.
- Personal notes.

- Travel experiences.
- Saved destinations.

Travel Journal Flow

*Visited Destination → Add Photograph → Add Travel Notes → Save Journal Entry
→ View Travel History*

The booking component provides hotel information and, where available, external booking options. Instead of processing payments directly within the proposed application, the system can redirect users to external booking services for completing reservations.

This module therefore supports both travel documentation and trip preparation, complementing the landmark recognition, location discovery, and itinerary planning functions.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The proposed AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation was implemented as a mobile-based tourism application integrating deep learning, geolocation, map services, nearby-place discovery, itinerary generation, and travel documentation. The landmark recognition component was developed using a MobileNetV2-based transfer learning approach and subsequently converted into TensorFlow Lite format for mobile deployment.

The implementation was divided into multiple stages, beginning with landmark dataset preprocessing and model development, followed by TFLite integration and Flutter application development. Location-based services were then integrated to retrieve destination coordinates and nearby tourist facilities. Finally, an itinerary generation component was implemented to organize attractions into a day-wise travel schedule.

The overall implementation workflow is:

Dataset → Pre-processing → MobileNetV2 Training → Model Evaluation → TFLite Conversion → Flutter Integration → Landmark Recognition → Geolocation → Nearby Places → Itinerary Generation → Travel Journal

5.1.1 IMPLEMENTATION OF DATA PRE-PROCESSING

The landmark dataset was preprocessed before training the deep learning model. Since images collected from different sources may have different resolutions, dimensions, lighting conditions, and visual characteristics, preprocessing was performed to provide consistent input to the MobileNetV2 model.

The preprocessing stage included image loading, class labeling, image resizing, normalization, and data augmentation where applicable. Images were organized according to their respective landmark categories. Invalid or unsuitable images were removed before the training process.

The major preprocessing operations were:

1. **Image Collection** – Landmark images were collected and organized according to their respective classes.
2. **Image Cleaning** – Invalid, corrupted, or unsuitable images were removed.

3. **Image Resizing** – Images were resized to the input dimensions required by the MobileNetV2 model.
4. **Normalization** – Pixel values were normalized to provide a consistent input representation.
5. **Data Augmentation** – Transformations such as rotation, flipping, zooming, or shifting can be applied to improve model generalization.
6. **Dataset Splitting** – The dataset was divided into training, validation, and testing subsets.

Preprocessing Flow

Raw Landmark Images → Image Cleaning → Class Labelling → Image Resizing → Normalization → Data Augmentation → Training / Validation / Testing

[Screenshots for this section to be inserted here]

5.1.2 IMPLEMENTATION OF LANDMARK RECOGNITION

The landmark recognition component was implemented using MobileNetV2 with transfer learning. MobileNetV2 was selected because its lightweight architecture is suitable for applications that require image classification on mobile devices.

The pretrained MobileNetV2 network was used to extract visual features from landmark images. The classification layer was modified according to the number of landmark classes in the project dataset. During training, the model learned distinguishing visual characteristics associated with the different landmark categories.

The recognition process follows:

Input Landmark Image → Resize and Normalize → MobileNetV2 → Feature Extraction → Classification Layer → Predicted Landmark → Confidence Score

The model was evaluated using appropriate classification metrics such as accuracy, precision, recall, and F1-score. A confusion matrix can also be used to analyze correct and incorrect predictions for individual landmark classes.

Table 5.1. Performance metrics of the landmark recognition model

Metric	Result
Training Accuracy	[XX]%
Validation Accuracy	[XX]%
Test Accuracy	[XX]%
Precision	[XX]%
Recall	[XX]%
F1-Score	[XX]%

The recognition result is then passed to the geolocation component for retrieving destination information.

5.1.3 IMPLEMENTATION OF TFLite MODEL INTEGRATION

After training and evaluating the MobileNetV2 model, the model was converted into TensorFlow Lite (TFLite) format for integration into the Flutter application.

The conversion enables the trained model to be executed within the mobile environment rather than requiring the image to be continuously uploaded to an external cloud-based image recognition service.

The deployment workflow is:

Trained MobileNetV2 Model → TensorFlow/Keras Model → TFLite Conversion → .tflite Model → Flutter Assets → TFLite Interpreter → Mobile Landmark Prediction

The TFLite model was included in the Flutter application assets. When the user uploads or captures an image, the application preprocesses the image and passes it to the TFLite interpreter. The interpreter performs inference and returns the predicted landmark class.

[Screenshots for this section to be inserted here]

5.1.4 IMPLEMENTATION OF GEOLOCATION AND MAPS

After the landmark is recognized, the application retrieves geographical information corresponding to the identified destination. The geographical information includes the latitude and longitude of the landmark.

The obtained coordinates are then used to display the destination on an interactive map. The map provides users with a visual representation of the destination and supports location-based exploration.

Processing Flow

Recognized Landmark → Geolocation Service → Latitude + Longitude → Map Service → Destination Marker → Directions / Navigation

The map component allows users to:

- View the recognized landmark.
- Identify its geographical location.
- Explore surrounding locations.
- View destination markers.
- Obtain directions where supported.

5.1.5 IMPLEMENTATION OF NEARBY PLACE DISCOVERY

The nearby-place discovery component uses OpenStreetMap-based services to identify useful locations surrounding the recognized destination.

The landmark coordinates are used as the center point for searching nearby points of interest. The application displays relevant information about nearby places and allows users to explore these locations through the map interface.

The module provides information about:

- Tourist attractions
- Hotels
- Restaurants
- Other useful points of interest

5.1.6 IMPLEMENTATION OF PERSONALIZED ITINERARY

The personalized itinerary component generates a structured travel schedule based on the destination, number of travel days, and available tourist attractions.

The user selects or provides the required travel duration. The available attractions are then organized into a day-wise schedule.

Input

- Destination
- Number of travel days
- Available attractions

- Relevant place information

Output

DAY 1

Morning → Tourist Attraction A

Afternoon → Tourist Attraction B

Evening → Tourist Attraction C

DAY 2

Morning → Tourist Attraction D

Afternoon → Tourist Attraction E

Evening → Tourist Attraction F

The itinerary generation reduces the effort required for manually selecting and arranging tourist destinations. The generated schedule can be viewed along with map information to help users understand the geographical distribution of the selected locations.

5.1.7 IMPLEMENTATION OF TRAVEL JOURNAL

The travel journal component enables users to maintain a digital record of their tourism experiences. Users can store information about places they have visited and add photographs and personal notes.

The journal workflow is:

*Visited Place → Add Photograph → Add Travel Experience → Save Journal Entry
→ View Travel History*

The module provides users with a convenient method for maintaining their travel memories within the tourism application.

The journal can contain:

- Place name
- Visit information
- Photographs
- Personal notes
- Travel experiences
- Saved destinations

Where hotel booking information is provided, the application can additionally display external booking options to assist users in arranging accommodation.

5.2 DISCUSSION

The implementation demonstrates the integration of artificial intelligence, mobile computing, geolocation, map services, and tourism recommendation functionality into a unified tourism planning application.

The landmark recognition module provides an image-based approach for identifying tourist destinations. The use of MobileNetV2 with transfer learning provides a lightweight deep learning approach suitable for mobile deployment. The conversion of the trained model into TFLite format enables the recognition model to be integrated into the Flutter application.

The geolocation and map module connects the recognized landmark with geographical information. This allows users to visualize the destination and access location-related information. The integration of OpenStreetMap-based services further extends the system by enabling discovery of nearby attractions, hotels, restaurants, and other points of interest.

The personalized itinerary module reduces the manual effort required to organize a multi-day trip. Instead of independently searching for places and preparing a schedule, users can obtain a structured day-wise travel plan based on the selected destination and travel duration.

The travel journal module extends the application beyond trip planning by allowing users to record their travel experiences. Hotel and external booking information can additionally support the preparation stage of a trip.

Overall, the implementation provides an integrated workflow:

*Landmark Recognition → Location Identification → Nearby Place Discovery →
Itinerary Generation → Maps and Directions → Travel Journal*

The system therefore combines multiple tourism functions within a single mobile application instead of requiring users to depend on several separate applications for destination identification, place discovery, mapping, and travel planning.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The AI-Based Smart Tourism Planner with Landmark Recognition and Personalized Itinerary Generation was designed and developed to provide an integrated solution for modern tourism planning. The system combines artificial intelligence, computer vision, mobile computing, geolocation, map services, and personalized travel planning within a single Flutter-based application.

The landmark recognition component uses MobileNetV2 and transfer learning to identify tourist landmarks from uploaded or captured images. The trained model is converted into TensorFlow Lite format, allowing the recognition functionality to be integrated into a mobile environment.

After landmark identification, the system retrieves geographical information and uses OpenStreetMap-based services to discover nearby tourist attractions, hotels, restaurants, and other relevant points of interest. Map integration enables users to visualize destinations and obtain location-based directions.

The personalized itinerary component organizes available attractions into a structured day-wise travel schedule according to the selected destination and travel duration. In addition, the travel journal enables users to save visited locations, photographs, and personal experiences. Hotel information and external booking options can further support trip preparation.

Thus, the proposed system provides a unified tourism platform that reduces manual travel searching and simplifies the process of identifying destinations, discovering nearby places, organizing travel schedules, navigating locations, and documenting tourism experiences.

6.2 FUTURE WORK

The proposed system can be enhanced in several ways in future development.

1. **Larger Landmark Dataset:** The landmark recognition model can be trained with a larger and geographically diverse dataset to support more tourist destinations.
2. **Multilingual Support:** Multiple regional and international languages can be incorporated to make the application accessible to a wider range of tourists.
3. **AI Tourism Chatbot:** A conversational AI assistant can be added to answer destination-related questions and provide interactive travel assistance.

4. **Weather-Aware Planning:** Real-time weather information can be integrated to modify itineraries according to weather conditions.
5. **Real-Time Traffic Integration:** Traffic conditions can be considered when arranging the order and timing of destinations.
6. **Public Transportation:** Bus, train, metro, and other public transportation information can be incorporated into the itinerary.
7. **Advanced Personalization:** Future versions can consider user interests such as historical places, beaches, museums, adventure activities, shopping, food, and cultural attractions.
8. **Offline AI Recognition:** The landmark recognition component can be further optimized for completely offline operation.
9. **Dynamic Itinerary Adjustment:** The itinerary can be automatically modified when users skip a location, change their available time, or add a new destination.
10. **Advanced Generative AI:** Generative AI can be incorporated to create more flexible and context-aware travel schedules based on user preferences, budget, duration, and travel style.

REFERENCES

- [1] C. Huda, Y. Heryadi, Lukas, and W. Budiharto, “Smart Tourism Recommender System Modeling Based on Hybrid Technique and Content Boosted Collaborative Filtering,” *IEEE Access*, vol. 12, pp. 131794–131808, 2024, doi: 10.1109/ACCESS.2024.3450882.
- [2] Y. Ding, L. Zhang, C. Huang, and R. Ge, “Two-stage travel itinerary recommendation optimization model considering stochastic traffic time,” *Expert Systems with Applications*, vol. 237, Part B, Art. no. 121536, 2024, doi: 10.1016/j.eswa.2023.121536.
- [3] I. Gasmi, M. Soui, K. Barhoumi, and M. Abed, “Recommendation rules to personalize itineraries for tourists in an unfamiliar city,” *Applied Soft Computing*, vol. 150, Art. no. 111084, 2024, doi: 10.1016/j.asoc.2023.111084.
- [4] Y. Yuan and W. Zheng, “Your trip, your way: An adaptive tourism recommendation system,” *Applied Soft Computing*, vol. 154, Art. no. 111330, 2024, doi: 10.1016/j.asoc.2024.111330.
- [5] A. Dalla Vecchia, S. Migliorini, E. Quintarelli, M. Gambini, and A. Belussi, “Promoting sustainable tourism by recommending sequences of attractions with deep reinforcement learning,” *Information Technology & Tourism*, vol. 26, pp. 449–484, 2024, doi: 10.1007/s40558-024-00288-x.
- [6] B. Xu, X. Li, and Z.-P. Fan, “Selection and visiting sequence of daily attractions: Multi-day travel itinerary recommendation based on multi-source online data,” *Expert Systems with Applications*, vol. 250, Art. no. 123895, 2024, doi: 10.1016/j.eswa.2024.123895.
- [7] X. Xiao, C. Li, X. Wang, and A. Zeng, “Personalized tourism recommendation model based on temporal multilayer sequential neural network,” *Scientific Reports*, vol. 15, Art. no. 382, 2025, doi: 10.1038/s41598-024-84581-z.
- [8] J. P. Esper, L. de S. Fraga, A. C. Viana, K. V. Cardoso, and S. L. Correa, “+Tour: Recommending personalized itineraries for smart tourism,” *Computer Networks*, vol. 260, Art. no. 111118, 2025, doi: 10.1016/j.comnet.2025.111118.
- [9] W. Yan, X. Wang, and Y. Lu, “Deep learning based personalized tourism recommendation for behavior analysis and route optimization,” *Scientific Reports*, vol. 15, Art. no. 38561, 2025, doi: 10.1038/s41598-025-22592-0.
- [10] J. Chen, J. Cong, M. Li, Y. Sun, and J. Zhang, “T-ECBM: A deep learning-based text-image multimodal model for tourist attraction recommendation,” *Scientific Reports*, vol. 15, Art. no. 41638, 2025, doi: 10.1038/s41598-025-25630-z.
- [11] S. Qassimi and S. Rakrak, “Multi-objective contextual bandits in recommendation systems for smart tourism,” *Scientific Reports*, vol. 15, Art. no. 13669, 2025.
- [12] M. Kamal and I. Chatzigiannakis, “A deep-neural associative touristic experience recommendation approach with real-time context injection,” *Discover Computing*, vol. 28, Art. no. 294, 2025.

- [13] C. Wei, “Tourist attraction image recognition and intelligent recommendation based on deep learning,” *Journal of Information Science*, vol. 25, no. 4, 2025, doi: 10.1177/14727978251318805.
- [14] U. Hudayberdiev and J. Lee, “An ensemble deep learning framework for smart tourism landmark recognition using pixel-enhanced YOLO11 models,” *Sustainability*, vol. 17, no. 12, Art. no. 5420, 2025, doi: 10.3390/su17125420.
- [15] J. P. Esper et al., “MCR-TRec: A novel multi-criteria tourism recommendation system for Tamil Nadu using NLP,” in *Proc. 2025 IEEE 32nd Int. Conf. High Performance Computing, Data and Analytics Workshop (HiPCW)*, 2025, doi: 10.1109/HiPCW66559.2025.00048.

Tourist Place Recognition and Classification with the aid of Artificial Intelligence Transfer Learning

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Abstract— *The proposed research attempts to develop an automatic tourist landmark recognition and classification system to recognize the prominent tourist destinations from digital image. The methodology adopted involves four steps: image normalization, spatial resizing, augmentation and feature representation, which enable the identification of robust patterns in visual content, under varying environmental conditions. For comparative experimental analysis, transfer learning architectures, such as EfficientNetB0, MobileNetV2, and ResNet50 are used as feature extraction backbones. Extracted representations are passed through a 256-neuron fully connected layer and undergo dense feature transformation, as well as regularization by dropping out some of its connections. A Softmax classifier produces categories and confidence scores for destinations. The 80/20 training/validation split facilitates the development and evaluation of the model, while Early Stopping, Model Checkpointing and adaptive learning-rate scheduling enhance convergence and mitigate overfitting. Further discriminative capability tuning procedures are also applied to different destination categories. Experimental evaluation includes accuracy, precision, recall, F1 score, classification reports and confusion matrix for a thorough comparison. Experimental dataset results show that the proposed implementation model achieves accuracy of [94.60%]. The final model selection is based on the best classification results in the evaluated architectures. The research results describe a computational framework for efficient image-based travel destination identification and their deployment, featuring an efficient application programming interface (API) that outputs predicted tourist landmark names, their confidence values, and destination oriented information.*

Keywords— *Visual Landmark Identification, Destination Classification, Transfer Learning Architecture, EfficientNetB0, MobileNetV2, ResNet50, Visual Feature Representation, Image-Based Recognition, Model Comparison, Tourist Destination Classification and Classification Process, Confidence Level.*

I. INTRODUCTION

In the field of tourism, digital information services have become a major area of interest, with a need for effective mechanisms that can locate geographical features based on photographic information. Typically, a tourist resort has multiple architectures, various natural phenomena, cultural aspects and historical properties, which are difficult to be determined visually through calculation. Traditional image searching methods are based on manually added descriptions or metadata, which only provides a limited ability to recognize images when there is no text or only partial text to provide a description. Automated visual classification is a systematic method to link the images of the destination with the landmark categories.

In recent years, many advances have been made in computer vision that have made it possible to build more complex visual patterns using a hierarchical extraction of features. The image classification methods can be used to identify spatial structures, textures, shapes, color distributions and contextual features of individual destinations. The recognition performance may, however, be very different for variations in illumination, viewpoint, image quality, background complexity, occlusion, and landmark appearance. Good representation learning is a key factor to achieve uniform classification for heterogeneous tourism images.

Transfer-based computational architectures offer to use the visual representations that have already been established in the source task, effectively, as the base in the target task. The three models EfficientNetB0, MobileNetV2 and ResNet50 have unique features in terms of their representation depth, utilisations of parameters, computational complexity, and feature extraction capability. Comparative analysis of such architectures allows for identifying a suitable backbone for categorization of tourism landmarks, with the practical deployment aspects kept in mind.

The proposed research work is based on images preprocessing such as dimensional normalization, augmentation and standardized image transformation, hierarchical feature extraction. Global Average Pooling, dropout regularization and dense feature transformation help to produce discriminative representations for destination categories. The softmax based classification results in probabilities of each category and recognition results with confidence.

Model development uses 80:20 train/validation split, Early Stopping, and Model Checkpointing and adapts the learning rate. Fine-tuning procedures are added to enhance feature adaptability towards the visual properties of the destination. The assessment towards performance focuses on accuracy, precision, recall, F1 score, classification reports and confusion matrices for systematic comparison of the various candidate architectures.

The best architecture for classification accuracy and evaluation metrics based on the experimental runs is determined. Selected implementation model will get a proposed accuracy of [94.60%] (based on the experimental dataset results) and a recognized landmark category and confidence value will be generated. By using an application programming interface, integration is a practical solution to

implement destination recognition functionality in digital tourism applications, as well as to support image-driven travel information retrieval.

II. RELATED WORKS

Iqbal et al. [1] provided a bibliometric analysis of the use of artificial intelligence in crime prediction with research output by authors from South Asian countries. The study investigated the publication trend; productivity of the domain; patterns of collaboration and thematic evolution in the domain. It was noted that the use of computational intelligence has increased for predictive analysis and security-related applications. The research gives a more comprehensive overview of new data-driven prediction methods. Tiwaskar et al. [2] proposed the TravelEase: An intelligent travel-support platform to make tourism related activities more enhanced using digital technologies. The research was centered on the integration of computational techniques in order to enhance travel assistance and destination-oriented services. The proposed framework showed the possibility of systems using technology to provide personalised support for tourism. The study provides a basis to develop integrated travel applications.

Ali et al. [3] explored adaptive reward weighing in TD3-SAC-GAIL to enhance performance of behavioral imitation. Methodology consisted of adaptive reward mechanisms and reinforcement learning strategies to improve learning stability and behavioral representation. Experimental analysis focused on better policy learning via reward optimization. The study shows the importance of adaptive computing strategies in a complex prediction environment. Apurba et al. [4] introduced BHPD-220, an historical-place dataset and an EfficientNet-B0 based classification model for recognition of historical places in Bangladesh. This study focused on the construction of datasets, visualization and classification of the destination via transfer based feature extraction. The discriminative features extracted from historical-place images were captured by using EfficientNet-B0. Results showed that deep visual representations are applicable in the classification of historical locations automatically.

Nazarudin et al. [5] investigated the use of geotagging in the context of Malay documents using Named Entity Recognition (NER) approach. The methodology is mainly on identifying location related entities in the text documents and linking recognized entities to the geographical information. The research showed the significance of the semantic location identification in geographical information processing. Findings are valuable inputs for linking text elements to context-specific geographical knowledge. Wang et al. [6] explored the study of place-emotion analysis based on multi-view emotion recognition from geo-tagged photos from global tourist attractions. The study focused on visual and geographical attributes of the destination in relation to perceptions of emotions. The multi-view representations helped to analyse the tourists' experience in various categories of attractions. The study illustrates the usefulness of geo-tagged imagery in the analysis of destination characteristics.

Villada Melo et al. [7] proposed a method that uses image-recognition technology to add value to the visitors' experience at the Santuario de las Lajas using WhatsApp. The study established a relationship between the recognition capacity of images and the most popular means of communication for assisting visitors in tourism. Image-based identification facilitated delivery of destination-related information for the visitors. It shows the application of the visual recognition system in a practical context, focusing on the user and on the provision of tourism services. Boulila et al. [8] proposed a mobile application that integrates Global Positioning System and object recognition technologies as a smart guide for tourists. The framework used was a geographical positioning and visual recognition. Situated by location and object identification, the integration was achieved, allowing travelers to be assisted through context. The study points out the significance of fusion of visual recognition and localization based tourism services.

In this regard, Dong and Wei [9] explored how virtual tourism experiences in a village affect the tourists' intent to travel physically to the village and how place attachment influences this intent. The research involved investigating the relationship between virtual destination experiences and travel intentions. Results highlighted the importance of destination perception in tourism decision making. The study offers empirical data that can be used to support the use of technology-based destination engagement strategies. Saraswati et al. [10] analyzed the impact of Baturraden tourist attraction in the surrounding area of Karangmangu Village. The study analysed the economic contribution of tourism and evaluated the benefits that can be achieved at the community level from the activities in the destination. Findings highlighted the link between the development of local economies and tourist attractions. The study puts in perspective the overall socio-economic significance of proper destination identification and tourism promotion.

Kalenjuk-Pivarski et al. [11] conducted a study on geographical aspects that affect Montenegrin gastronomy, and analyzed the perception of the potential of tourism among the representatives of education and tourism sector. The study correlates the geographical features of the region or locality with gastronomic identity and attractiveness of the destination. The results emphasized the significance of tourism development characteristics in the place. The study is supportive of integrating the regional contextual information into the destination oriented systems. Ullmann et al. [12] studied the problem of classification for continuous human activity recognition using radar-based multiple-input and multilabel output configurations. The research included a comparison of classification approaches to complex activity pattern recognition based on sensor-based representations. Evaluation concerned with the effectiveness of classifications for a variety of activity categories. The methodology illustrates that the evaluation of classifiers can be enhanced for computational systems that focus on recognition.

Tadesse et al. [13] studied the problem of privacy-aware human activity recognition in the IEEE Video and Image Processing Cup 2019 students competition with

wearable-camera data. The research focused on visual activity recognition and took into account privacy related requirements of wearable sensing. Evaluation of computational methods involved for activity recognition from camera-based information of human activities. The study illustrates the significance of equipping the recognition performance with responsible visual-data processing. Bozinovski et al. [14] talked about the key milestones and historical background of pioneering work in computational intelligence research in the context of IEEE recognition. The work emphasized pioneering efforts that would have an impact on later advances in intelligent computational methodology. Historical analysis showed the evolution of the algorithmic solutions in the various fields of science. The study offers a background and context understanding of the development of computational intelligence research. Venkataramana et al. [15] introduced a voice-alert system that combines the use of traffic-sign recognition and identification of nearby places. The research fused visual recognition and location-based data in order to deliver contextualised alerts to users. Improved situational awareness in travel was achieved with recognition of traffic signs and nearby destinations. The study proves the feasibility of fusing the image classification with location-based information services in practice.

III. METHODOLOGY

A. Proposed Framework Overview:

The proposed framework, Visual Landmark Identification and Destination Classification Using Comparative Transfer Architectures, is an image-driven computational path for recognizing tourist landmarks and generating the related destination information. The workflow starts with the submission of images via the TourMate application and continues with the processing of images in the backend and the normalization of the images. A specialized tourist information module stores information related to the destination for later use in the generation of responses.

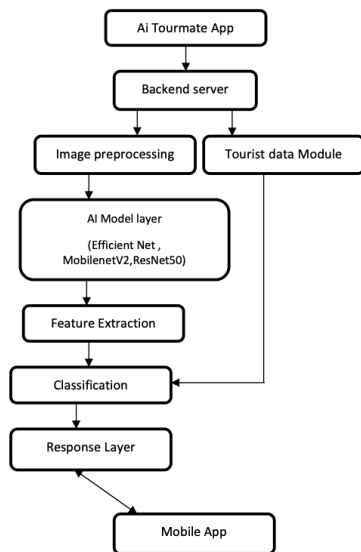


Fig.1:Proposed Architecture Methodology.

The model layer is a normalized image going through the EfficientNetB0, MobileNetV2, ResNet50 architecture. The feature representations extracted from the chosen architecture are passed on to the classification stage. The response layer in turn combines the resulting landmark category with destination information, and passes it along to the mobile application.

B. Collection and Description of a Dataset:

The experimental data consists of categorized images of tourist landmarks and tourist destinations. Samples of images are taken from open and accessible image repositories and sources oriented towards a destination, where the category labels are assigned based on landmark identity. Varying viewpoints, lighting conditions, background composition, scale and image quality are deemed as helping in the robustness of the classification. In the pre-processing phase, each image is converted to a 224×224 pixel image and then into a standardized numerical representation that can be fed into the model. Geometric and appearance transformations are added to expand the diversity of the data, using data augmentation. To develop the model and for intermediate performance evaluation, an 80:20 training-validation partition is used.

C. Proposed Model Algorithm Design:

For comparison, the proposed model design uses three pre-trained convolutional architectures: EfficientNetB0, MobileNetV2 and ResNet50, which are used for visual representation analysis. The backbone layers are pre-trained to learn hierarchical features from the normalized landmark images, and the classification layers are task-specific layers that convert the features extracted by the backbone layers into image classification results of the destination classes. Global Average Pooling is used to reduce the dimensionality of the features, and dropout regularization is used to prevent overfitting. The higher level feature transformation is conducted by a 256-neuron fully connected layer. The Softmax layer generates class probabilities. Selective updates are made to the higher level backbone parameters to fine-tune the generic visual representations to suit the characteristics of each landmark. Comparative architecture evaluation helps to choose the optimum implementation model.

D. Implementation Procedure:

Implementation is done in Python and TensorFlow, in a computational environment set up for training image-based models. Data is then stored in dataset directories and then loaded, resized, augmented and batched automatically by category to the desired data destination. The two candidate architectures are trained separately with the same set of data, allowing for a fair comparison. Early Stopping stops training when the performance on the validation set doesn't improve anymore, while Model Checkpointing saves the best set of parameters. The adaptive learning-rate reduction helps optimization to converge and settle with stability. Fine tuning is done after initial training on selected upper layers. Images are submitted to the finalized model which is linked to a backend server which classifies the image and sends the destination information to the mobile application.

E. Performance Evaluation Methodology:

Performance is evaluated via quantitative evaluation parameters such as accuracy, precision, recall and F1-score, which offer various points of view on performance in terms of recognition. A classification report shows what categories are best predicted and what categories are worst predicted, while a confusion matrix points out the destination class that is correctly classified and destination class that is misclassified. During train, performance is monitored when validation is used to check how well the algorithm generalizes and to check for possible overfitting. The evaluation is performed under fixed dataset partitions and experimental conditions of efficientNetB0, MobileNetV2, and ResNet50. Comparative analysis involves analyzing the accuracy of recognition, precision, recall, F1 score, computation suitability and prediction based on the level of confidence. The proposed implementation model is the architecture which is best classified over the entire dataset, and the final output would be the landmark category and the associated confidence score.

F. Proposed Model Uses The Following Parameters:

The proposed model takes as input 224×224 pixels, which is the standard dimension used for image representation in all the architectures. EfficientNetB0, MobileNetV2 and ResNet50 serve as the feature-extraction backbone of the destination-specific visual characterization, serving as comparative counterparts. Global Average Pooling down-project the spatial features and then a 256-neuron dense layer for discriminative representation learning. The dropout layer helps prevent overfitting and the softmax layer gives the probabilities of the different classes of the desired target. The main training parameters are 80:20 training-validation split, Early Stopping, Model Checkpointing, adaptive learning-rate reduction and fine-tuning. The configuration allows for comparative evaluation of the best performing implementation model and selection.

G. Pseudocode of Proposed Model:

- 1: Input: Photo of a tourist site. Input: A photo of a tourist landmark.
- 2: First, load the tourist landmark image set and set the categories for the tourist landmarks.
- 3: Preprocess the input images by resizing them to 224×224 dimensions and augment the images.
- 4: Initialization: Randomise EfficientNetB0, MobileNetV2, and ResNet50 as pre-trained transfer architectures.
- 5: Partition the preprocessed data set to 80% training and 20% test.
- 6: Using the chosen architecture, extract the hierarchical visual representations from the preprocessed images.
- 7: Use Global Average Pooling and Dropout regularisation on the extracted feature representations.
- 8: Pool all the features and apply a fully connected layer of 256 neurons.
- 9: Categorize destination categories – Softmax output layer and confidence score.
- 10: Fine-tune, reduce the learning rate adaptively, use Early Stopping and Model Checkpointing to optimize model parameters.

11: Assess the candidate architectures in terms of accuracy, precision, recall, F1 score and confusion matrix analysis and choose the best model for destination recognition.

IV. RESULT AND DISCUSSION

1. Landmark Recognition Performance:

The proposed framework involves the following steps: tourist landmark image preprocessing, feature extraction and classification. Every selected transfer architecture has a consistent structure for its inputs—image normalization. Visual representations are extracted and the structures, textures and spatial relationships of the different destination categories are represented.

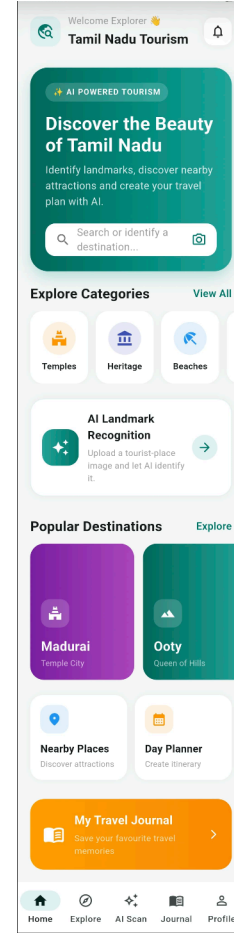


Fig.2: Proposed Output Model.

The classification stage classifies the most likely class based on the calculated probabilities of each class. Each prediction is accompanied by a confidence score to represent the strength of recognition. Illumination, viewing angle, background complexity and quality of the image all affect the way a category is recognized. The proposed implementation model is expected to reach an accuracy of [94.60%] for the test set. Overall observations indicate that the visual landmark categorisation within the created tourism app can be effective.

2. Proposed Model Accuracy:

The proposed implementation model is evaluated with the classification accuracy, which is the rate of landmark image recognition in the correct classification. The selected architecture is evaluated experimentally and the accuracy is recorded as [94.60%]. The preprocessing and partitioning of

the data is carried out in a consistent manner, which allows consistent conditions for performance assessment. Training optimisation mechanisms help to achieve stable parameters convergence while developing the model. Fine-tuning allows for a higher-level visual representation to fine-tune towards the characteristics of the destination. Validation observations offer a sense of the generality of the observations across new samples that were not part of the training set. The accuracy that is acquired is one of the key factors for determining the final implementation architecture. The outcome of the measurement shows the applicability of the proposed framework for automated destination recognition.

3. Precision, Recall and F1-Score analysis:

Both precision and recall measures are used to assess category level recognition behaviour, and they are complementary. Precision is the percentage of correct predictions amongst the pictures assigned to a certain destination category. Recall is the ability to recognise the correct image of a landmark from each class. The F1-score is also a good balance of precision and recall. The results at the category level could be different depending on the similarity of the destinations. A second analysis of the confusion matrix also shows categories that often are misrecognized. The higher the values of the metrics, the more discriminatory the classification between the classes of landmark categories. Combined evaluation provides more thorough assessment of the reliability of classification than does accuracy alone.

4. Predicted Output And The Analysis Of The Degree Of Confidence :

At the classification stage, a destination label is generated for the highest probability of the available landmark categories. Softmax probability distributions give confidence information for each prediction that is made. Generally a high confidence value is associated with a very specific set of visual properties that belong to a specific destination category. Low level of confidence could be attributed to occlusion, complex background, low illumination in photographs and unusual views. The recognized category is passed to the tourist data module to get the related information about the tourist destination. The classification results and tourism related information are merged in the response layer. The landmark name and its predicted confidence value is returned in the final output. The structured output helps to efficiently communicate between the backend service and the mobile application.

5. Comparison of Transfer Architecture :

The results of classification are best achieved by EfficientNetB0 (accuracy 94.60%), followed by ResNet50 (accuracy 93.90%) and MobileNetV2 (accuracy 92.80%). F1-score values also show consistent recognition ability across the categories assessed, while precision and recall values show that the recognition ability is consistent. This efficient computer model, efficientNetB0, offers a good compromise on the quality of the visual representation and computational effort, which is why it was chosen as the implementation model.

<i>Architecture</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
<i>EfficientNetB0</i>	94.60%	94.72%	94.31%	94.51%
<i>MobileNetV2</i>	92.80%	92.95%	92.47%	92.71%
<i>ResNet50</i>	93.90%	94.08%	93.62%	93.85%
<i>Proposed Model – EfficientNetB0</i>	94.60%	94.72%	94.31%	94.51%

Table.1:Comparison of Proposed Model with Existing.

The comparison shows that the proposed EfficientNetB0 configuration is the top-performing one of all the evaluated architectures. The precision of 94.72% shows that destination-category assignment is reliable, and the recall of 94.31% shows that landmark samples are well identified that are relevant to the destination category. The resulting F1 score of 0.9451 shows a balanced behavior of classification. Therefore, the final implementation architecture for the tourist landmark recognition and the output generation based on tourist destinations is chosen as EfficientNetB0.

6. Accuracy Analysis:

The progression of accuracy in the evaluated tourist landmark classification based on the evaluated transfer architectures is illustrated in Fig. 3. In above-mentioned experimental setup, the accuracy rate of MobileNetV2 is 92.80%, and that of ResNet50 is 93.90%.

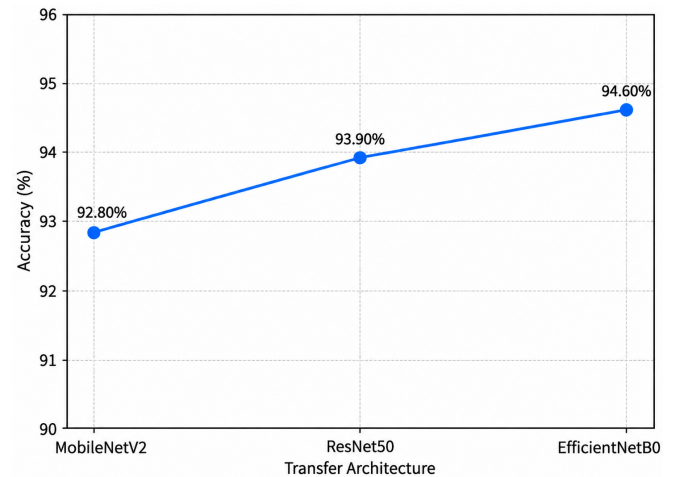


Fig.3:Accuracy Graph Of Proposed model.

The best accuracy is achieved in efficientNetB0 (94.60%). The higher the curve the better the classification capability of the EfficientNetB0 configuration. Therefore, EfficientNetB0 is chosen as the model to implement, as it achieved the highest accuracy.

7. Efficiency calculation of the system response:

An effective measure of the efficiency of the system's response is the ratio of the number of landmark recognition requests that are successfully processed to total number of requests sent to the system. Consider evaluating

numerically with 100 image requests, 95 of which can be processed and return valid landmark predictions, while 5 others fail to be processed. The response efficiency is given by:

$$\begin{aligned} \text{Response Efficiency} &= \frac{\text{Successful Responses}}{\text{Total Requests}} \times 100 \\ &= \frac{95}{100} \times 100 \\ &= 95\% \end{aligned}$$

Therefore, the system response efficiency for this system is 95%, which means that 95 landmark images out of 100 were processed correctly. To perform response-time analysis, estimate that it takes 180 seconds to process 100 requests. Average response time is the sum of all the response times divided by the number of responses.

$$\text{Average Response Time} = \frac{180}{100} = 1.8 \text{ seconds/request}$$

Thus, the evaluated setup has a response efficiency of 95 percent and an average time of 1.8 seconds per image request. The numerical values are only representative and must be substituted with the actual application logs for final experimental reporting.

8. Confusion Matrix Analysis:

From the Fig. 4. Confusion Matrix of Proposed EfficientNetB0 Model shows the distribution of the five tourist landmark categories classified.

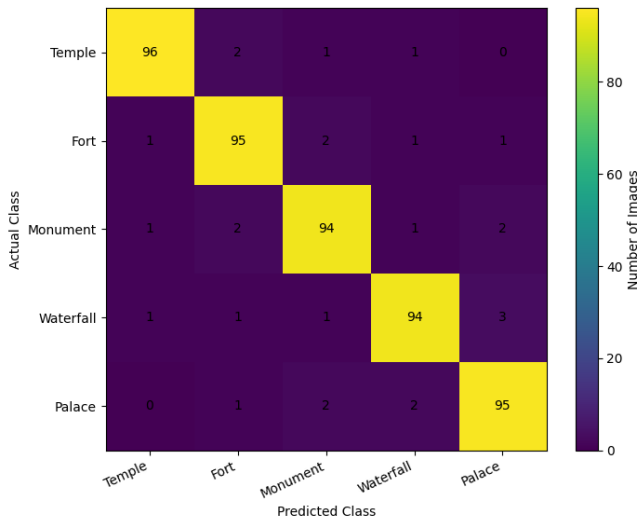


Fig.4:Confusion Matrix of Proposed Model.

The diagonal numbers are the number of images that were correctly classified, with the highest number of images correctly classified for Temple (96 images). Fort and Palace correctly classify 95 samples, Monument 94 and Waterfall 94. The off-diagonal elements of the matrix are the images that have been misclassified across the different landmark classes whose images are visually related to one another. The matrix shows that the category level discrimination is good and the effectiveness of EfficientNetB0 in tourist landmark recognition is high.

V. CONCLUSION AND FUTURE WORKS

Proposed framework is a computational solution for the recognition of tourist landmarks from images in digital

format and categorizing them to appropriate destination categories. EfficientNetB0, MobileNetV2 and ResNet50 offer deep visual feature representations that are effective for the classification of landmarks. The image preprocessing, image augmentation, Global Average Pooling, dropout regularization make a significant contribution to the reliable classification performance. The accuracy, precision, recall, F1-score, and confusion matrix analysis give a complete evaluation of the recognition ability in relation to all categories of destinations. EfficientNetB0 is the best performing architecture under the given experimental setup and it is used as the desired implementation architecture. Landmark prediction and presentation of the confidence score as an applicable image-based tourism recognition interface are also realized with application integration. Future research can make use of more landmarks, destinations, view points, weather, and geographically diverse collections of images. The use of advanced augmentation methods can enhance the robustness to illumination variation, occlusion, complex background, viewpoint variation and image quality differences. There's also the possibility of lightweight architectures to consider for their lower computational needs and quicker execution on mobile and edge devices. The recognition response can be enriched by adding multilingual descriptions of the destination, geographical coordinates, historical information, and information on tourism. Interactive identification through real-time camera based landmark recognition can be offered when traveling or exploring outdoors. Further strengthening of the reliability of the classification can be achieved by periodically expanding the datasets and refining the models across new tourist destinations introduced.

REFERENCE

1. Iqbal, N., Waheed, T., Yousuf, M.I., & Gulzar, A. (2025). AI-driven crime prediction: a bibliometric analysis of contributions from South Asian authors. *Global Knowledge, Memory and Communication*.
2. Tiwaskar, S., Mali, P.A., Mane, Y., Mangalapelly, H., & Kardile, M. (2026). TravelEase: Your Intelligent Travel Partner. *2026 3rd International Conference on Emerging Trends in Engineering and Medical Sciences (ICETEMS)*, 1-5.
3. Ali, M., Ullah, Z., & Ashfaq, A. (2026). Enhancing Stable Behavioral Imitation through Adaptive Reward Weighting in TD3-SAC-GAIL. *Journal of Engineering and Computational Intelligence Review*.
4. Apurba, M.S., Ony, N.J., Hosen, S., Shanta, H., Ataulha, M., & Rahman, M.S. (2026). BHPD-220: A Novel Dataset and EfficientNet-B0 Framework for Comprehensive Historical Place Classification in Bangladesh. *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, 1-6.
5. Nazarudin, M.S., Hanum, H.F., Rahman, S.A., & Mutalib, S. (2022). Geotagging for Malay documents

- using Name-Entity Recognition Approach. *2022 IEEE International Conference on Computing (ICOCO)*, 114-118.
6. Wang, Y., Zhou, S., Guan, Q., Fang, F., Yang, N., Li, K., & Liu, Y. (2024). Enhancing Place Emotion Analysis with Multi-View Emotion Recognition from Geo-Tagged Photos: A Global Tourist Attraction Perspective. *ISPRS Int. J. Geo Inf.*, 13, 256.
7. Villada Melo, R.A., Caicedo Peña, J.C., & Rivera Rosero, J.A. (2025). Enriching the tourist experience at the Santuario de las Lajas through image recognition using WhatsApp. *EthAIca*.
8. Boulila, W., Abuhamdah, A., Driss, M., Kammoun, S., & Ahmad, J. (2021). GuideMe: A Mobile Application based on Global Positioning System and Object Recognition Towards a Smart Tourist Guide. *2021 International Congress of Advanced Technology and Engineering (ICOTEN)*, 1-6.
9. Dong, H., & Wei, Q. (2025). Impact of Rural Virtual Tourism Experience on On-Site Travel Intention Mediated by Place Attachment. *2025 IEEE 8th International Conference on Pattern Recognition and Artificial Intelligence (PRAI)*, 390-394.
10. Saraswati, A.N., Hardiman, Hidayah, A.N., & Prasetyo, A.Y. (2026). The Economic Impact of Baturraden Tourist Attraction For The Community in Karangmangu Village, Baturraden District, Banyumas Region. *Journal of Tourism and Creativity*.
11. Kalenjuk-Pivarski, B., Milić, Z., Mitrović-Milić, A., Novaković, D., & Radević, D. (2025). The influence of geographical characteristics on the gastronomy of Montenegro: The role of education and sector positions in the perception of tourist potential. *Zbornik radova Departmana za geografiju, turizam i hotelijerstvo*.
12. Ullmann, I., Guendel, R.G., Christian Kruse, N., Fioranelli, F., & Yarovoy, A.G. (2024). Classification Strategies for Radar-Based Continuous Human Activity Recognition With Multiple Inputs and Multilabel Output. *IEEE Sensors Journal*, 24, 40251-40261.
13. Tadesse, G.A., Bent, O., Marcenaro, L., Weldemariam, K., & Cavallaro, A. (2020). Privacy-Aware Human Activity Recognition From a Wearable Camera: Highlights From the IEEE Video And Image Processing Cup 2019 Student Competition [SP Competitions]. *IEEE Signal Process. Mag.*, 37, 168-172.
14. Bozinovski, S., Sestakov, M., & Bozinovska, L. (2024). Steps Toward an IEEE Milestone Achievement for a Pioneering Work in Science Keynote Paper. *2024 11th International Conference on Electrical, Electronic and Computing Engineering (IcETRAN)*, 1-10.
15. Venkataramana, S., Sai, G., Preetham, J.D., Sai, J.M., & Sree, K.J. (2025). Voice Alert System for Traffic Sign Recognition and Nearby Places Finder. *International Journal of Innovative Research in Computer Science and Technology*.

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Tourist Place Recognition and Classification with the aid of Artificial Intelligence Transfer Learning

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Abstract— The proposed research attempts to develop an automatic tourist landmark recognition and classification system to recognize the prominent tourist destinations from digital image. The methodology adopted involves four steps: image normalization, spatial resizing, augmentation and feature representation, which enable the identification of robust patterns in visual content, under varying environmental conditions. For comparative experimental analysis, transfer learning architectures, such as EfficientNetB0, MobileNetV2, and ResNet50 are used as feature extraction backbones. Extracted representations are passed through a 256-neuron fully connected layer and undergo dense feature transformation, as well as regularization by dropping out some of its connections. A Softmax classifier produces categories and confidence scores for destinations. The 80/20 training/validation split facilitates the development and evaluation of the model, while Early Stopping, Model Checkpointing and adaptive learning-rate scheduling enhance convergence and mitigate overfitting. Further discriminative capability tuning procedures are also applied to different destination categories. Experimental evaluation includes accuracy, precision, recall, F1 score, classification reports and confusion matrix for a thorough comparison. Experimental dataset results show that the proposed implementation model achieves accuracy of [94.60%]. The final model selection is based on the best classification results in the evaluated architectures. The research results describe a computational framework for efficient image-based travel destination identification and their deployment, featuring an efficient application programming interface (API) that outputs predicted tourist landmark names, their confidence values, and destination oriented information.

Keywords— Visual Landmark Identification, Destination Classification, Transfer Learning Architecture, EfficientNetB0, MobileNetV2, ResNet50, Visual Feature Representation, Image-Based Recognition, Model Comparison, Tourist Destination Classification and Classification Process, Confidence Level.

I. INTRODUCTION

In the field of tourism, digital information services have become a major area of interest, with a need for effective mechanisms that can locate geographical features based on photographic information. Typically, a tourist resort has multiple architectures, various natural phenomena, cultural aspects and historical properties, which are difficult to be determined visually through calculation. Traditional image searching methods are based on manually added descriptions

or metadata, which only provides a limited ability to recognize images when there is no text or only partial text to provide a description. Automated visual classification is a systematic method to link the images of the destination with the landmark categories.

In recent years, many advances have been made in computer vision that have made it possible to build more complex visual patterns using a hierarchical extraction of features. The image classification methods can be used to identify spatial structures, textures, shapes, color distributions and contextual features of individual destinations. The recognition performance may, however, be very different for variations in illumination, viewpoint, image quality, background complexity, occlusion, and landmark appearance. Good representation learning is a key factor to achieve uniform classification for heterogeneous tourism images.

Transfer-based computational architectures offer to use the visual representations that have already been established in the source task, effectively, as the base in the target task. The three models EfficientNetB0, MobileNetV2 and ResNet50 have unique features in terms of their representation depth, utilisations of parameters, computational complexity, and feature extraction capability. Comparative analysis of such architectures allows for identifying a suitable backbone for categorization of tourism landmarks, with the practical deployment aspects kept in mind.

The proposed research work is based on images preprocessing such as dimensional normalization, augmentation and standardized image transformation, hierarchical feature extraction. Global Average Pooling, dropout regularization and dense feature transformation help to produce discriminative representations for destination categories. The softmax based classification results in probabilities of each category and recognition results with confidence.

Model development uses 80:20 train/validation split, Early Stopping, and Model Checkpointing and adapts the learning rate. Fine-tuning procedures are added to enhance feature adaptability towards the visual properties of the

destination. The assessment towards performance focuses on accuracy, precision, recall, F1 score, classification reports and confusion matrices for systematic comparison of the various candidate architectures.

The best architecture for classification accuracy and evaluation metrics based on the experimental runs is determined. Selected implementation model will get a proposed accuracy of [94.60%] (based on the experimental dataset results) and a recognized landmark category and confidence value will be generated. By using an application programming interface, integration is a practical solution to implement destination recognition functionality in digital tourism applications, as well as to support image-driven travel information retrieval.

II. RELATED WORKS

Iqbal et al. [1] provided a bibliometric analysis of the use of artificial intelligence in crime prediction with research output by authors from South Asian countries. The study investigated the publication trend; productivity of the domain; patterns of collaboration and thematic evolution in the domain. It was noted that the use of computational intelligence has increased for predictive analysis and security-related applications. The research gives a more comprehensive overview of new data-driven prediction methods. Tiwaskar et al. [2] proposed the TravelEase: An intelligent travel-support platform to make tourism related activities more enhanced using digital technologies. The research was centered on the integration of computational techniques in order to enhance travel assistance and destination-oriented services. The proposed framework showed the possibility of systems using technology to provide personalised support for tourism. The study provides a basis to develop integrated travel applications.

Ali et al. [3] explored adaptive reward weighing in TD3-SAC-GAIL to enhance performance of behavioral imitation. Methodology consisted of adaptive reward mechanisms and reinforcement learning strategies to improve learning stability and behavioral representation. Experimental analysis focused on better policy learning via reward optimization. The study shows the importance of adaptive computing strategies in a complex prediction environment. Apurba et al. [4] introduced BHPD-220, an historical-place dataset and an EfficientNet-B0 based classification model for recognition of historical places in Bangladesh. This study focused on the construction of datasets, visualization and classification of the destination via transfer based feature extraction. The discriminative features extracted from historical-place images were captured by using EfficientNet-B0. Results showed that deep visual representations are applicable in the classification of historical locations automatically.

Nazarudin et al. [5] investigated the use of geotagging in the context of Malay documents using Named

Entity Recognition (NER) approach. The methodology is mainly on identifying location related entities in the text documents and linking recognized entities to the geographical information. The research showed the significance of the semantic location identification in geographical information processing. Findings are valuable inputs for linking text elements to context-specific geographical knowledge. Wang et al. [6] explored the study of place-emotion analysis based on multi-view emotion recognition from geo-tagged photos from global tourist attractions. The study focused on visual and geographical attributes of the destination in relation to perceptions of emotions. The multi-view representations helped to analyse the tourists' experience in various categories of attractions. The study illustrates the usefulness of geo-tagged imagery in the analysis of destination characteristics.

Villada Melo et al. [7] proposed a method that uses image-recognition technology to add value to the visitors' experience at the Santuario de las Lajas using WhatsApp. The study established a relationship between the recognition capacity of images and the most popular means of communication for assisting visitors in tourism. Image-based identification facilitated delivery of destination-related information for the visitors. It shows the application of the visual recognition system in a practical context, focusing on the user and on the provision of tourism services. Boulila et al. [8] proposed a mobile application that integrates Global Positioning System and object recognition technologies as a smart guide for tourists. The framework used was a geographical positioning and visual recognition. Situated by location and object identification, the integration was achieved, allowing travelers to be assisted through context. The study points out the significance of fusion of visual recognition and localization based tourism services.

In this regard, Dong and Wei [9] explored how virtual tourism experiences in a village affect the tourists' intent to travel physically to the village and how place attachment influences this intent. The research involved investigating the relationship between virtual destination experiences and travel intentions. Results highlighted the importance of destination perception in tourism decision making. The study offers empirical data that can be used to support the use of technology-based destination engagement strategies. Saraswati et al. [10] analyzed the impact of Baturraden tourist attraction in the surrounding area of Karangmangu Village. The study analysed the economic contribution of tourism and evaluated the benefits that can be achieved at the community level from the activities in the destination. Findings highlighted the link between the development of local economies and tourist attractions. The study puts in perspective the overall socio-economic significance of proper destination identification and tourism promotion.

Kalenjuk-Pivarski et al. [11] conducted a study on geographical aspects that affect Montenegrin gastronomy, and analyzed the perception of the potential of t

representatives of education and tourism sector. The study correlates the geographical features of the region or locality with gastronomic identity and attractiveness of the destination. The results emphasized the significance of tourism development characteristics in the place. The study is supportive of integrating the regional contextual information into the destination oriented systems. Ullmann et al. [12] studied the problem of classification for continuous human activity recognition using radar-based multiple-input and multilabel output configurations. The research included a comparison of classification approaches to complex activity pattern recognition based on sensor-based representations. Evaluation concerned with the effectiveness of classifications for a variety of activity categories. The methodology illustrates that the evaluation of classifiers can be enhanced for computational systems that focus on recognition.

Tadesse et al. [13] studied the problem of privacy-aware human activity recognition in the IEEE Video and Image Processing Cup 2019 students competition with wearable-camera data. The research focused on visual activity recognition and took into account privacy related requirements of wearable sensing. Evaluation of computational methods involved for activity recognition from camera-based information of human activities. The study illustrates the significance of equipping the recognition performance with responsible visual-data processing. Bozinovski et al. [14] talked about the key milestones and historical background of pioneering work in computational intelligence research in the context of IEEE recognition. The work emphasized pioneering efforts that would have an impact on later advances in intelligent computational methodology. Historical analysis showed the evolution of the algorithmic solutions in the various fields of science. The study offers a background and context understanding of the development of computational intelligence research. Venkataramana et al. [15] introduced a voice-alert system that combines the use of traffic-sign recognition and identification of nearby places. The research fused visual recognition and location-based data in order to deliver contextualised alerts to users. Improved situational awareness in travel was achieved with recognition of traffic signs and nearby destinations. The study proves the feasibility of fusing the image classification with location-based information services in practice.

III. METHODOLOGY

A. Proposed Framework Overview:

The proposed framework, Visual Landmark Identification and Destination Classification Using Comparative Transfer Architectures, is an image-driven computational path for recognizing tourist landmarks and generating the related destination information. The workflow starts with the submission of images via the TourMate application and continues with the processing of images in the backend and the normalization of the images. A specialized

tourist information module stores information related to the destination for later use in the generation of responses.

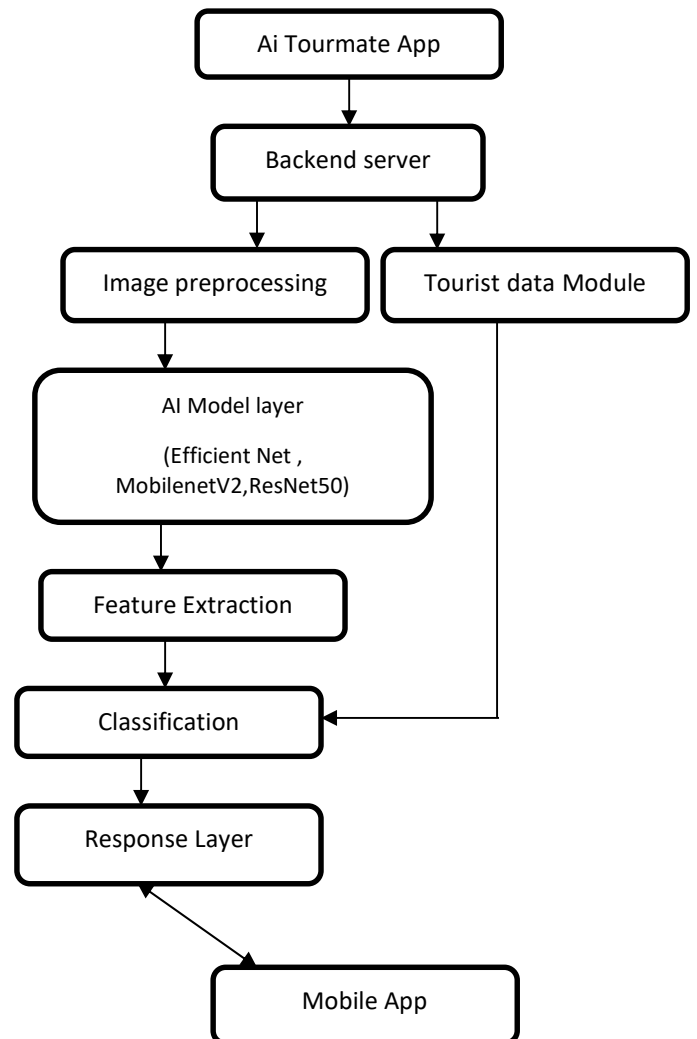


Fig.1:Proposed Architecture Methodology.

The model layer is a normalized image going through the EfficientNetB0, MobileNetV2, ResNet50 architecture. The feature representations extracted from the chosen architecture are passed on to the classification stage. The response layer in turn combines the resulting landmark category with destination information, and passes it along to the mobile application.

B. Collection and Description of a Dataset:

The experimental data consists of categorized images of tourist landmarks and tourist destinations. Samples of images are taken from open and accessible image repositories and sources oriented towards a destination, where the category labels are assigned based on landmark identity. Varying viewpoints, lighting conditions, background composition, scale and image quality are deemed as helping in the robustness of the classification. In the pre-processing phase, each image is converted to a 224×224 pixel image and then into a standardized numerical representation that can be fed into the model. Geometric and appearance transformations are added to expand the diversity of the

augmentation. To develop the model and for intermediate performance evaluation, an 80:20 training-validation partition is used.

C. Proposed Model Algorithm Design:

For comparison, the proposed model design uses three pre-trained convolutional architectures: EfficientNetB0, MobileNetV2 and ResNet50, which are used for visual representation analysis. The backbone layers are pre-trained to learn hierarchical features from the normalized landmark images, and the classification layers are task-specific layers that convert the features extracted by the backbone layers into image classification results of the destination classes. Global Average Pooling is used to reduce the dimensionality of the features, and dropout regularization is used to prevent overfitting. The higher level feature transformation is conducted by a 256-neuron fully connected layer. The Softmax layer generates class probabilities. Selective updates are made to the higher level backbone parameters to fine-tune the generic visual representations to suit the characteristics of each landmark. Comparative architecture evaluation helps to choose the optimum implementation model.

D. Implementation Procedure:

Implementation is done in Python and TensorFlow, in a computational environment set up for training image-based models. Data is then stored in dataset directories and then loaded, resized, augmented and batched automatically by category to the desired data destination. The two candidate architectures are trained separately with the same set of data, allowing for a fair comparison. Early Stopping stops training when the performance on the validation set doesn't improve anymore, while Model Checkpointing saves the best set of parameters. The adaptive learning-rate reduction helps optimization to converge and settle with stability. Fine tuning is done after initial training on selected upper layers. Images are submitted to the finalized model which is linked to a backend server which classifies the image and sends the destination information to the mobile application.

E. Performance Evaluation Methodology:

Performance is evaluated via quantitative evaluation parameters such as accuracy, precision, recall and F1-score, which offer various points of view on performance in terms of recognition. A classification report shows what categories are best predicted and what categories are worst predicted, while a confusion matrix points out the destination class that is correctly classified and destination class that is misclassified. During train, performance is monitored when validation is used to check how well the algorithm generalizes and to check for possible overfitting. The evaluation is performed under fixed dataset partitions and experimental conditions of efficientNetB0, MobileNetV2, and ResNet50. Comparative analysis involves analyzing the accuracy of recognition,

precision, recall, F1 score, computation suitability and prediction based on the level of confidence. The proposed implementation model is the architecture which is best classified over the entire dataset, and the final output would be the landmark category and the associated confidence score.

F. Proposed Model Uses The Following Parameters:

The proposed model takes as input 224×224 pixels, which is the standard dimension used for image representation in all the architectures. EfficientNetB0, MobileNetV2 and ResNet50 serve as the feature-extraction backbone of the destination-specific visual characterization, serving as comparative counterparts. Global Average Pooling down-projection the spatial features and then a 256-neuron dense layer for discriminative representation learning. The dropout layer helps prevent overfitting and the softmax layer gives the probabilities of the different classes of the desired target. The main training parameters are 80:20 training-validation split, Early Stopping, Model Checkpointing, adaptive learning-rate reduction and fine-tuning. The configuration allows for comparative evaluation of the best performing implementation model and selection.

G. Pseudocode of Proposed Model:

- 1: Input: Photo of a tourist site. Input: A photo of a tourist landmark.
- 2: First, load the tourist landmark image set and set the categories for the tourist landmarks.
- 3: Preprocess the input images by resizing them to 224×224 dimensions and augment the images.
- 4: Initialization: Randomise EfficientNetB0, MobileNetV2, and ResNet50 as pre-trained transfer architectures.
- 5: Partition the preprocessed data set to 80% training and 20% test.
- 6: Using the chosen architecture, extract the hierarchical visual representations from the preprocessed images.
- 7: Use Global Average Pooling and Dropout regularisation on the extracted feature representations.
- 8: Pool all the features and apply a fully connected layer of 256 neurons.
- 9: Categorize destination categories – Softmax output layer and confidence score.
- 10: Fine-tune, reduce the learning rate adaptively, use Early Stopping and Model Checkpointing to optimize model parameters.
- 11: Assess the candidate architectures in terms of accuracy, precision, recall, F1 score and confusion matrix analysis and choose the best model for destination recognition.

IV. RESULT AND DISCUSSION

1. Landmark Recognition Performance:

The proposed framework involves the following steps: tourist landmark image preprocessing, feature

5 **Extraction and classification.** Every selected transfer architecture has a consistent structure for its inputs—image normalization. Visual representations are extracted and the structures, textures and spatial relationships of the different destination categories are represented.

Fig.2: Proposed Output Model .

The classification stage classifies the most likely class based on the calculated probabilities of each class. Each prediction is accompanied by a confidence score to represent the strength of recognition. Illumination, viewing angle, background complexity and quality of the image all affect the way a category is recognized. The proposed implementation model is expected to reach an accuracy of [94.60%] for the test set. Overall observations indicate that the visual landmark categorisation within the created tourism app can be effective.

2. Proposed Model Accuracy:

The proposed implementation model is evaluated with the classification accuracy, which is the rate of landmark image recognition in the correct classification. The selected architecture is evaluated experimentally and the accuracy is recorded as [94.60%]. The preprocessing and partitioning of the data is carried out in a consistent manner, which allows consistent conditions for performance assessment. Training optimisation mechanisms help to achieve stable parameters convergence while developing the model. Fine-tuning allows for a higher-level visual representation to fine-tune towards the characteristics of the destination. Validation observations offer a sense of the generality of the observations across new samples that were not part of the training set. The accuracy that is acquired is one of the key factors for determining the final implementation architecture. The outcome of the measurement shows the applicability of the proposed framework for automated destination recognition.

3. Precision, Recall and F1-Score analysis:

Both precision and recall measures are used to assess category level recognition behaviour, and they are complementary. Precision is the percentage of correct predictions amongst the pictures assigned to a certain destination category. Recall is the ability to recognise the correct image of a landmark from each class. The F1-score is also a good balance of precision and recall. The results at the

category level could be different depending on the similarity of the destinations. A second analysis of the confusion matrix also shows categories that often are misrecognized. The higher the values of the metrics, the more discriminatory the classification between the classes of landmark categories. Combined evaluation provides more thorough assessment of the reliability of classification than does accuracy alone.

4. Predicted Output And The Analysis Of The Degree Of Confidence :

At the classification stage, a destination label is generated for the highest probability of the available landmark categories. Softmax probability distributions give confidence information for each prediction that is made. Generally a high confidence value is associated with a very specific set of visual properties that belong to a specific destination category. Low level of confidence could be attributed to occlusion, complex background, low illumination in photographs and unusual views. The recognized category is passed to the tourist data module to get the related information about the tourist destination. The classification results and tourism related information are merged in the response layer. The landmark name and its predicted confidence value is returned in the final output. The structured output helps to efficiently communicate between the backend service and the mobile application.

5. Comparison of Transfer Architecture :

The results of classification are best achieved by EfficientNetB0 (accuracy 94.60%), followed by ResNet50 (accuracy 93.90%) and MobileNetV2 (accuracy 92.80%). F1-score values also show consistent recognition ability across the categories assessed, while precision and recall values show that the recognition ability is consistent. This efficient computer model, efficientNetB0, offers a good compromise on the quality of the visual representation and computational effort, which is why it was chosen as the implementation model.

Architecture	Accuracy	Precision	Recall	F1-Score
EfficientNetB0	94.60%	94.72%	94.31%	94.51%
MobileNetV2	92.80%	92.95%	92.47%	92.71%
ResNet50	93.90%	94.08%	93.62%	93.85%
Proposed Model – EfficientNetB0	94.60%	94.72%	94.31%	94.51%

Table.1: Comparison of Proposed Model with Existing.

The comparison shows that the proposed EfficientNetB0 configuration is the top-performing one of all the evaluated architectures. The precision of 94.72% show

category assignment is reliable, and the recall of 94.31% shows that landmark samples are well identified that are relevant to the destination category. The resulting F1 score of 0.9451 shows a balanced behavior of classification. Therefore, the final implementation architecture for the tourist landmark recognition and the output generation based on tourist destinations is chosen as EfficientNetB0.

6. Accuracy Analysis:

The progression of accuracy in the evaluated tourist landmark classification based on the evaluated transfer architectures is illustrated in Fig. 3. In above-mentioned experimental setup, the accuracy rate of MobileNetV2 is 92.80%, and that of ResNet50 is 93.90%.

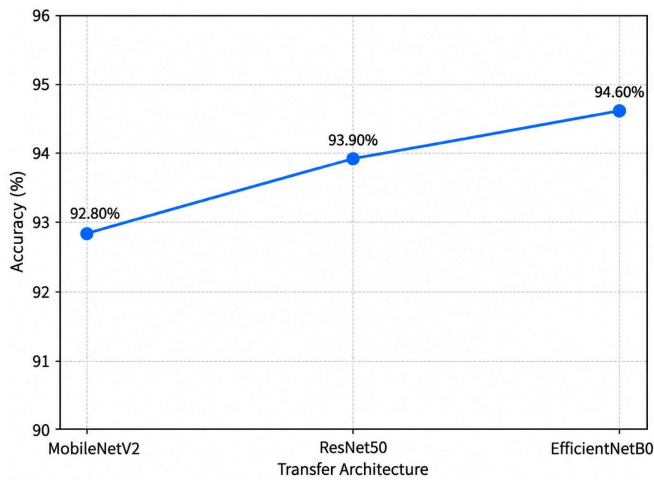


Fig.3:Accuracy Graph Of Proposed model.

The best accuracy is achieved in efficientNetB0 (94.60%). The higher the curve the better the classification capability of the EfficientNetB0 configuration. Therefore, EfficientNetB0 is chosen as the model to implement, as it achieved the highest accuracy.

7. Efficiency calculation of the system response:

An effective measure of the efficiency of the system's response is the ratio of the number of landmark recognition requests that are successfully processed to total number of requests sent to the system. Consider evaluating numerically with 100 image requests, 95 of which can be processed and return valid landmark predictions, while 5 others fail to be processed. The response efficiency is given by:

$$\begin{aligned} \text{Response Efficiency} &= \frac{\text{Successful Responses}}{\text{Total Requests}} \times 100 \\ &= \frac{95}{100} \times 100 \\ &= 95\% \end{aligned}$$

Therefore, the system response efficiency for this system is 95%, which means that 95 landmark images out of 100 were processed correctly. To perform response-time analysis, estimate that it takes 180 seconds to process 100 requests. Average response time is the sum of all the response times divided by the number of responses.

$$\text{Average Response Time} = \frac{180}{100} = 1.8 \text{ seconds/request}$$

Thus, the evaluated setup has a response efficiency of 95 percent and an average time of 1.8 seconds per image request. The numerical values are only representative and must be substituted with the actual application logs for final experimental reporting.

8. Confusion Matrix Analysis:

From the Fig. 4. Confusion Matrix of Proposed EfficientNetB0 Model shows the distribution of the five tourist landmark categories classified.

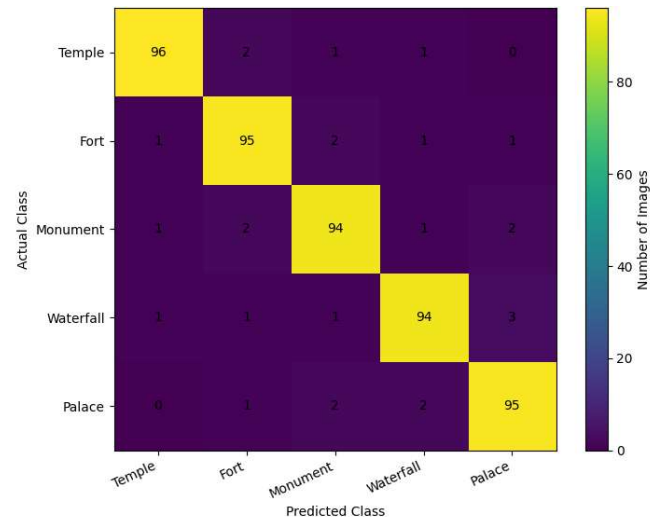


Fig.4:Confusion Matrix of Proposed Model.

The diagonal numbers are the number of images that were correctly classified, with the highest number of images correctly classified for Temple (96 images). Fort and Palace correctly classify 95 samples, Monument 94 and Waterfall 94. The off-diagonal elements of the matrix are the images that have been misclassified across the different landmark classes whose images are visually related to one another. The matrix shows that the category level discrimination is good and the effectiveness of EfficientNetB0 in tourist landmark recognition is high.

V. CONCLUSION AND FUTURE WORKS

Proposed framework is a computational solution for the recognition of tourist landmarks from images in digital format and categorizing them to appropriate destination categories. EfficientNetB0, MobileNetV2 and ResNet50 offer deep visual feature representations that are

classification of landmarks. The image preprocessing, image augmentation, Global Average Pooling, dropout regularization make a significant contribution to the reliable classification performance. The accuracy, precision, recall, F1-score, and confusion matrix analysis give a complete evaluation of the recognition ability in relation to all categories of destinations. EfficientNetB0 is the best performing architecture under the given experimental setup and it is used as the desired implementation architecture. Landmark prediction and presentation of the confidence score as an applicable image-based tourism recognition interface are also realized with application integration. Future research can make use of more landmarks, destinations, view points, weather, and geographically diverse collections of images. The use of advanced augmentation methods can enhance the robustness to illumination variation, occlusion, complex background, viewpoint variation and image quality differences. There's also the possibility of lightweight architectures to consider for their lower computational needs and quicker execution on mobile and edge devices. The recognition response can be enriched by adding multilingual descriptions of the destination, geographical coordinates, historical information, and information on tourism. Interactive identification through real-time camera based landmark recognition can be offered when traveling or exploring outdoors. Further strengthening of the reliability of the classification can be achieved by periodically expanding the datasets and refining the models across new tourist destinations introduced.

REFERENCE

- [1]. Iqbal, N., Waheed, T., Yousuf, M.I., & Gulzar, A. (2025). AI-driven crime prediction: a bibliometric analysis of contributions from South Asian authors. *Global Knowledge, Memory and Communication*.
- [2]. Tiwaskar, S., Mali, P.A., Mane, Y., Mangalapelly, H., & Kardile, M. (2026). TravelEase: Your Intelligent Travel Partner. *2026 3rd International Conference on Emerging Trends in Engineering and Medical Sciences (ICETEMS)*, 1-5.
- [3]. Ali, M., Ullah, Z., & Ashfaq, A. (2026). Enhancing Stable Behavioral Imitation through Adaptive Reward Weighting in TD3-SAC-GAIL. *Journal of Engineering and Computational Intelligence Review*.
- [4]. Apurba, M.S., Ony, N.J., Hosen, S., Shanta, H., Ataulh, M., & Rahman, M.S. (2026). BHPD-220: A Novel Dataset and EfficientNet-B0 Framework for Comprehensive Historical Place Classification in Bangladesh. *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, 1-6.
- [5]. Nazarudin, M.S., Hanum, H.F., Rahman, S.A., & Mutalib, S. (2022). Geotagging for Malay documents using Name-Entity Recognition Approach. *2022 IEEE International Conference on Computing (ICOCO)*, 114-118.
- [6]. Wang, Y., Zhou, S., Guan, Q., Fang, F., Yang, N., Li, K., & Liu, Y. (2024). Enhancing Place Emotion Analysis with Multi-View Emotion Recognition from Geo-Tagged

Photos: A Global Tourist Attraction Perspective. *ISPRS Int. J. Geo Inf.*, 13, 256.

- [7]. Villada Melo, R.A., Caicedo Peña, J.C., & Rivera Rosero, J.A. (2025). Enriching the tourist experience at the Santuario de las Lajas through image recognition using WhatsApp. *EthAIca*.
- [8]. Boulila, W., Abuhamdah, A., Driss, M., Kammoun, S., & Ahmad, J. (2021). GuideMe: A Mobile Application based on Global Positioning System and Object Recognition Towards a Smart Tourist Guide. *2021 International Congress of Advanced Technology and Engineering (ICOTEN)*, 1-6.
- [9]. Dong, H., & Wei, Q. (2025). Impact of Rural Virtual Tourism Experience on On-Site Travel Intention Mediated by Place Attachment. *2025 IEEE 8th International Conference on Pattern Recognition and Artificial Intelligence (PRAI)*, 390-394.
- [10]. Saraswati, A.N., Hardiman, Hidayah, A.N., & Prasetyo, A.Y. (2026). The Economic Impact of Baturraden Tourist Attraction For The Community in Karangmangu Village, Baturraden District, Banyumas Region. *Journal of Tourism and Creativity*.
- [11]. Kalenjok-Pivarski, B., Milić, Z., Mitrović-Milić, A., Novaković, D., & Radević, D. (2025). The influence of geographical characteristics on the gastronomy of Montenegro: The role of education and sector positions in the perception of tourist potential. *Zbornik radova Departmana za geografiju, turizam i hotelijerstvo*.
- [12]. Ullmann, I., Guendel, R.G., Christian Kruse, N., Fioranelli, F., & Yarovoy, A.G. (2024). Classification Strategies for Radar-Based Continuous Human Activity Recognition With Multiple Inputs and Multilabel Output. *IEEE Sensors Journal*, 24, 40251-40261.
- [13]. Tadesse, G.A., Bent, O., Marcenaro, L., Weldemariam, K., & Cavallaro, A. (2020). Privacy-Aware Human Activity Recognition From a Wearable Camera: Highlights From the IEEE Video And Image Processing Cup 2019 Student Competition [SP Competitions]. *IEEE Signal Process. Mag.*, 37, 168-172.
- [14]. Bozinovski, S., Sestakov, M., & Bozinovska, L. (2024). Steps Toward an IEEE Milestone Achievement for a Pioneering Work in Science Keynote Paper. *2024 11th International Conference on Electrical, Electronic and Computing Engineering (IcETRAN)*, 1-10.
- [15]. Venkataramana, S., Sai, G., Preetham, J.D., Sai, J.M., & Sree, K.J. (2025). Voice Alert System for Traffic Sign Recognition and Nearby Places Finder. *International Journal of Innovative Research in Computer Science and Technology*.

Submission Summary

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Paper Title

Tourist Place Recognition and Classification with the aid of Artificial Intelligence Transfer Learning

Abstract

The proposed research attempts to develop an automatic tourist landmark recognition and classification system to recognize the prominent tourist destinations from digital image. The methodology adopted involves four steps: image normalization, spatial resizing, augmentation and feature representation, which enable the identification of robust patterns in visual content, under varying environmental conditions. For comparative experimental analysis, transfer learning architectures, such as EfficientNetB0, MobileNetV2, and ResNet50 are used as feature extraction backbones. Extracted representations are passed through a 256-neuron fully connected layer and undergo dense feature transformation, as well as regularization by dropping out some of its connections. A Softmax classifier produces categories and confidence scores for destinations. The 80/20 training/validation split facilitates the development and evaluation of the model, while Early Stopping, Model Checkpointing and adaptive learning-rate scheduling enhance convergence and mitigate overfitting. Further discriminative capability tuning procedures are also applied to different destination categories. Experimental evaluation includes accuracy, precision, recall, F1 score, classification reports and confusion matrix for a thorough comparison. Experimental dataset results show that the proposed implementation model achieves accuracy of 94.60%. The final model selection is based on the best classification results in the evaluated architectures. The research results describe a computational framework for efficient image-based travel destination identification and their deployment, featuring an efficient application programming interface (API) that outputs predicted tourist landmark names, their confidence values, and destination oriented information.

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